

Selenium Testing Guide

1. Selenium Testing Objectives

- Verify web application functionality through browser automation.
- Verify page navigation, forms, buttons, links, and user interactions.
- Verify application behavior across supported browsers.
- Verify validation messages and error handling.

2. Login Automation Scenarios

- Verify login with valid username and password.
- Verify login with an invalid password.
- Verify login with an invalid username.
- Verify login when required fields are empty.
- Verify successful navigation to the dashboard after login.
- Verify logout ends the authenticated session.

3. UI Element Testing

- Locate buttons, text fields, links, dropdowns, checkboxes, and radio buttons.
- Verify elements are visible and enabled before interaction.
- Verify expected text and attributes.
- Verify validation messages appear for invalid input.

4. Test Execution

- Open the application in the configured browser.
- Navigate to the required page.
- Perform the test steps using Selenium.
- Capture the actual result.
- Compare the actual result with the expected result.
- Record pass or fail status.

5. Cross-Browser Testing

- Run important test cases in Chrome.
- Run important test cases in Edge.
- Run important test cases in Firefox.
- Verify consistent behavior across supported browsers.

6. Common Selenium Checks

- Verify page title and URL.
- Verify element visibility.
- Verify element click behavior.
- Verify text input and clearing.
- Verify navigation and redirects.
- Verify alerts and browser dialogs where applicable.

7. Regression Testing

- Execute critical automated test cases after application changes.
- Verify previously fixed defects.
- Verify important login and navigation workflows.
- Record failed tests and relevant evidence.